

BAD702: SMD MODULE 3

Question 1

Describe the concept of P-value in hypothesis testing. Why is it important and how is it interpreted?

Answer

1. Definition

A **P-value (probability value)** is the probability of obtaining a test statistic **as extreme as, or more extreme than, the observed result**, assuming that the **null hypothesis H_0** is true.

It is used to determine how strongly the sample evidence contradicts the null hypothesis.

$$\text{P-value} = P(\text{observed or more extreme result} \mid H_0 \text{ is true})$$

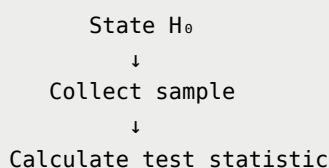
2. Role of P-value in Hypothesis Testing

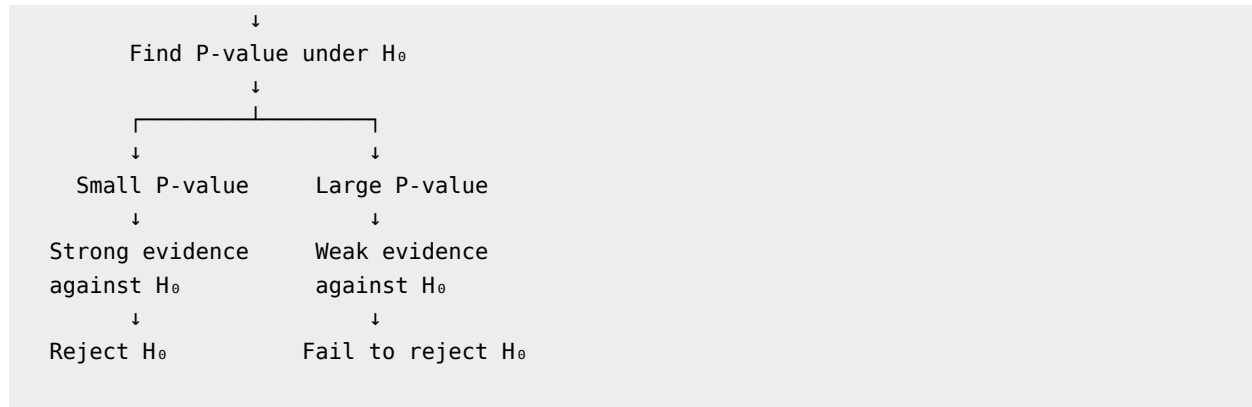
Hypothesis testing generally involves two hypotheses:

- **Null hypothesis (H_0)**: There is no significant effect, difference, or relationship.
- **Alternative hypothesis (H_1)**: There is a significant effect, difference, or relationship.

The P-value is calculated from the observed test statistic and the **sampling distribution under H_0** .

Diagram





3. Interpretation of P-value

Small P-value

A **small P-value** indicates that the observed result would be relatively unusual if H_0 were true.

Therefore:

Small P-value $\Rightarrow$ strong evidence against H_0

For example:

$$p=0.01$$

means that, assuming H_0 is true, a result at least as extreme as the observed result has probability about **1%**.

This provides strong evidence against H_0 .

Large P-value

A **large P-value** indicates that the observed result is reasonably compatible with H_0 .

Large P-value $\Rightarrow$ weak evidence against H_0

It does **not** prove that H_0 .

4. Decision Using Significance Level

A significance level α is selected before performing the test, commonly:
 $\alpha = 0.05$

The P-value is compared with α .

Condition	Decision	Interpretation
$p \leq \alpha$	Reject H0	Significant evidence against H0
$p > \alpha$	Fail to reject H0	Insufficient evidence against H0

Example

Suppose:

$$\alpha = 0.05$$

and the calculated P-value is:

$$p = 0.02$$

Since:

$$0.02 < 0.05$$

we **reject H0**.

There is statistically significant evidence against the null hypothesis.

5. Why is the P-value Important?

1. Measures strength of evidence

The P-value provides a quantitative measure of how unusual the observed result is under the null hypothesis.

2. Helps make decisions

It provides a systematic basis for deciding whether to **reject or fail to reject H0**.

3. Used in statistical inference

P-values are widely used in **hypothesis tests** involving means, proportions, regression coefficients, categorical data and other statistical models.

4. Indicates statistical significance

When:

$$p \leq \alpha$$

the result is considered **statistically significant** at significance level α .

6. Important Points About P-value

The following interpretations should be avoided:

- A P-value is **not** the probability that H_0 is true.
- A large P-value does **not prove** H_0 .
- A small P-value does not necessarily mean the observed effect is **practically important**.
- The P-value depends on the **sample size** and magnitude of the observed effect.

Thus, statistical significance should be considered together with the **effect size, confidence interval and practical importance** of the result.

7. Summary

$$p \leq \alpha \Rightarrow \text{Reject } H_0$$

$$p > \alpha \Rightarrow \text{Fail to reject } H_0$$

The smaller the P-value, the stronger the evidence against the null hypothesis, assuming the statistical model and test assumptions are appropriate.

Question 2:

Write the step-by-step procedure for performing a two-sample t-test.

Answer

1. Introduction

A **two-sample t-test** is used to determine whether there is a statistically significant difference between the **means of two groups**.

It is commonly used for comparing two groups in an **A/B test** when the data is numeric.

For example, it can be used to compare the average response time for **Page A and Page B**.

2. Step-by-Step Procedure

Step 1: State the hypotheses

Define the **null hypothesis H_0** and **alternative hypothesis H_1** .

For comparing two population means:

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

For a one-sided test, the alternative hypothesis may be:

$$H_1 : \mu_1 < \mu_2$$

or

$$H_1 : \mu_1 > \mu_2$$

Step 2: Collect the two samples

Obtain two independent samples:

$$X_1, X_2, \dots, X_{n_1}$$

and

$$Y_1, Y_2, \dots, Y_{n_2}$$

where:

- n_1, n_2 = sample sizes
- $\bar{x}_1, \bar{x}_2$ = sample means
- s_1, s_2 = sample standard deviations.

Step 3: Calculate the difference between sample means

Calculate:

$$D = \bar{x}_1 - \bar{x}_2$$

This is the observed difference between the two groups.

Step 4: Calculate the standard error

For the commonly used **Welch two-sample t-test**, the standard error is:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Welch's test does not require the two population variances to be equal.

Step 5: Calculate the t-statistic

The test statistic is:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)_0}{SE}$$

Usually, under the null hypothesis:

$$(\mu_1 - \mu_2)_0 = 0$$

Therefore:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The t -statistic is a **standardized measure of the observed difference**.

Step 6: Determine the reference distribution

The calculated t -statistic is compared with the appropriate **t -distribution**.

For Welch's test, the degrees of freedom are approximated using the Welch–Satterthwaite method:

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

The t -distribution is used as a reference distribution for differences between two sample means.

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Step 7: Calculate the P-value

The **P-value** indicates how unusual the observed difference would be if the null hypothesis were true.

For a two-sided test:

$$p = P(|T| \geq |t_{\text{observed}}| \mid H_0)$$

Step 8: Compare with significance level

Choose a significance level, commonly:

$$\alpha = 0.05$$

Then:

Then:

$$p \leq \alpha \Rightarrow \text{Reject } H_0$$

$$p > \alpha \Rightarrow \text{Fail to reject } H_0$$

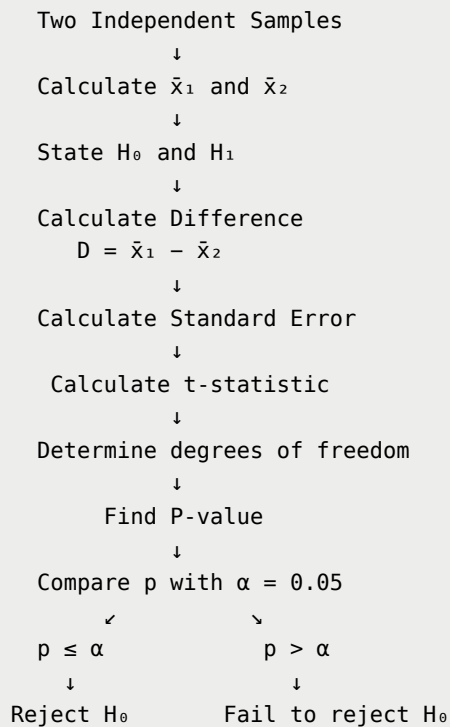
Step 9: State the conclusion

Interpret the result in the **context of the problem**.

- If $p \leq 0.05$: there is statistically significant evidence that the two population means differ.
- If $p > 0.05$: there is insufficient evidence to conclude that the population means differ.

Do **not** say that a large P-value proves that the two means are equal.

3. Flow Diagram



4. Example

Suppose two web pages produce average session times:

$$\bar{x}_A = 126.33$$

$$\bar{x}_B = 162.00$$

The objective is to test whether Page A has a **smaller mean session time** than Page B.

Set:

$$H_0 : \mu_A = \mu_B$$

$$H_1 : \mu_A < \mu_B$$

The t-test calculates the standardized difference and its corresponding P-value. In the example from the statistical analysis procedure, the reported result was:

$$t = -1.0983, \quad df = 27.693, \quad p = 0.1408$$

Since the P-value is greater than 0.05, the null hypothesis is **not rejected** at the 5% significance level.

5. Important Points

- A two-sample t-test compares **two sample means**.
- The **t-statistic** standardizes the difference between the means.
- The **t-distribution** is used as the reference distribution.
- **Degrees of freedom** are required to determine the appropriate t-distribution.
- The **P-value** is used to assess evidence against H_0 .
- **Welch's two-sample t-test** is commonly used when equal population variances should not be assumed.

Question 3:

What is multiple testing problem? How does it affect statistical inference and how is it corrected?

Answer

1. Definition

The **multiple testing problem** occurs when **multiple hypotheses or statistical comparisons are tested using the same data**. As the number of tests increases, the probability of obtaining at least one **statistically significant result purely by chance** also increases.

This problem is also called **alpha inflation**.

2. Why Does Multiple Testing Cause a Problem?

Suppose each individual hypothesis test is performed at:

$$\alpha = 0.05$$

For one test, there is a **5% probability of making a Type I error** when the null hypothesis is true.

However, when many independent tests are performed, the probability that **at least one test is falsely significant** becomes much larger.

For n independent tests:

$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^n$$

Example

If 20 independent tests are performed at $\alpha = 0.05$:

$$\begin{aligned} P(\text{at least one false positive}) &= 1 - (0.95)^{20} \\ &\approx 0.642 \end{aligned}$$

Thus, there is approximately a **64% chance of obtaining at least one false significant result**, even when all null hypotheses are true.

This increase in the overall Type I error probability is called **alpha inflation**.

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3. Effect on Statistical Inference

Multiple testing can lead to **false discoveries** and misleading conclusions.

The problem becomes more serious when researchers:

- Perform many pairwise comparisons.
- Examine many subgroups.
- Try many statistical models.
- Include many variables in models.
- Ask many different questions about the same data.

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For example, with three treatment groups A, B, C , we may perform three comparisons:

A vs B

B vs C

A vs C

Each additional test increases the chance of finding something **significant** simply because of **random variation**.  BAD702-module-3-textbook(1)

This can result in:

Multiple tests → Alpha inflation → More false positives → Misleading inference

It is closely related to **overfitting**, where repeated searching or modeling effectively results in **fitting the noise** rather than finding a genuine effect.  BAD702-module-3-textbook(1)

4. Correction of Multiple Testing

The problem can be addressed using **statistical adjustment procedures**.

The basic idea is to make the criterion for significance **more stringent** when multiple tests are performed.

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A. Bonferroni Adjustment

The simplest method is the **Bonferroni adjustment**.

If:

- Overall significance level = α
- Number of tests/comparisons = m

then the significance level for each individual test is:

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m}$$

Example

Suppose:

and 5 hypotheses are tested.

Then:

$$\alpha_{\text{adjusted}} = \frac{0.05}{5} = 0.01$$

Therefore, each individual test must have:

$$p \leq 0.01$$

to be considered statistically significant.

Thus, Bonferroni **divides up alpha according to the number of tests**, providing a more stringent significance threshold.  BAD702-module-3-textbook(1)

5. Tukey's HSD

Another procedure is **Tukey's Honest Significant Difference (HSD)**.

It is particularly useful when **comparing multiple group means**.

Tukey's HSD considers the **maximum difference among group means** and compares it with a benchmark based on the **t-distribution**.  BAD702-module-3-textbook(1)

It is therefore appropriate for structured multiple comparisons such as:

$$A \text{ vs } B, \quad B \text{ vs } C, \quad A \text{ vs } C$$

6. Holdout Sample in Data Mining

In **data mining and predictive modeling**, a **holdout sample** can help reduce the risk of reaching misleading conclusions from repeated analysis.

7. Summary

Aspect	Explanation
Multiple testing	Performing many hypothesis tests/comparisons
Main problem	Increased probability of false positives
Type of error	Type I error
Alpha inflation	Increase in overall Type I error probability
Effect	False discoveries and misleading statistical inference

Aspect	Explanation
Bonferroni	α / m for each test
Tukey HSD	Adjusts comparisons among multiple group means
Holdout sample	Helps avoid misleading findings in predictive modeling

Question 4:

What is the role of degrees of freedom in t-tests and other inferential statistics?

Answer

1. Definition

Degrees of freedom (d.f.) represent the number of independent pieces of information available for estimating a parameter after accounting for the **constraints** imposed by the calculation.

In inferential statistics, degrees of freedom are an important part of **standardizing test statistics** so that they can be compared with the appropriate **reference distribution**, such as the **t-distribution** or **F-distribution**.

2. Degrees of Freedom in a Sample

For a sample of size n , when estimating the sample variance or standard deviation, the degrees of freedom are:

$$df = n - 1$$

Why $n - 1$?

When the sample mean $\bar{x}$ has been calculated, the deviations from the mean must satisfy:

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

Therefore, once $n - 1$ deviations are known, the final deviation is **constrained** and can be determined from the others.

Hence:

$$\text{Degrees of freedom} = n - 1$$

The $n - 1$ degrees of freedom account for the **one constraint** introduced by estimating the sample mean.

3. Role in t-tests

The **t-distribution** is used as a reference distribution for:

- Sample means
- Differences between two sample means
- Regression parameters
- Other statistics. BAD702-module-2-textbook(1)

For a one-sample t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

The corresponding degrees of freedom are:

$$df = n - 1$$

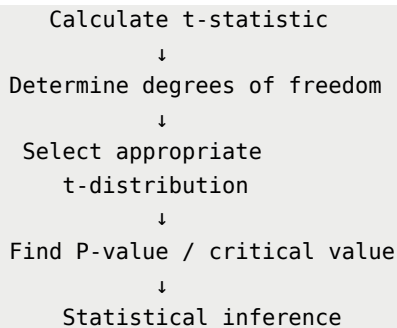
Thus, the degrees of freedom determine **which t-distribution** should be used to evaluate the calculated t-statistic.

4. Effect on the t-distribution

The t-distribution is actually a **family of distributions**, each corresponding to a different number of degrees of freedom.

Sample Data

↓



For **small degrees of freedom**, the t-distribution has **thicker tails** than the normal distribution.

As the degrees of freedom increase, the t-distribution becomes increasingly similar to the **normal distribution**.

5. Role in Confidence Intervals

Degrees of freedom are also used when constructing confidence intervals using the t-distribution.

For example, a confidence interval for a sample mean is:

$$\bar{x} \pm t_{df, \alpha/2} \frac{s}{\sqrt{n}}$$

For a sample of size n :

$$df = n - 1$$

where $t_{df, \alpha/2}$ is the appropriate critical value from the t-distribution.

For example, a **90% confidence interval** uses the t-value that leaves 5% of the distribution in each tail.

6. Degrees of Freedom in Other Inferential Statistics

Degrees of freedom are not limited to t-tests.

A. F-tests

Degrees of freedom are required for the **F-distribution**, particularly in:

- **ANOVA**
- Regression analysis
- Comparison of variances.

The appropriate F-distribution depends on its degrees of freedom.

B. Chi-square Tests

Degrees of freedom are also associated with the **Chi-square distribution** and are used when performing tests involving categorical data and other statistical procedures.

Thus, degrees of freedom help match a standardized statistic to its correct **reference distribution**.

7. Degrees of Freedom in Regression

Degrees of freedom are also important when dealing with **categorical variables in regression**.

Suppose a categorical variable has n categories. It requires:

$$n - 1$$

indicator or **dummy variables**, rather than n , to avoid perfect redundancy and **multicollinearity**.

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Example

For **7 days of the week**:

$$7 - 1 = 6$$

degrees of freedom are available.

Therefore, six dummy variables can be used, with the seventh category represented by the reference category.

8. Importance of Degrees of Freedom

The main roles are:

1. **Select the appropriate reference distribution.**
2. Determine the appropriate **critical value** for hypothesis testing.
3. Help calculate **P-values**.
4. Determine the appropriate width of **confidence intervals**.
5. Account for **constraints** introduced while estimating parameters.
6. Help avoid redundancy when representing categorical variables in **regression models**.

9. Summary Table

Application	Role of Degrees of Freedom
One-sample t-test	$df = n - 1$
t-based confidence interval	Determines appropriate t critical value
Two-sample t-test	Determines appropriate t reference distribution
F-test / ANOVA	Determines the appropriate F-distribution
Chi-square test	Determines the appropriate Chi-square distribution
Regression	Helps determine parameter/model degrees of freedom
Categorical variables	$n - 1$ dummy variables avoid redundancy

Question 5:

Explain A/B testing in data science. Discuss its design, randomization, treatment groups, control groups and interpretation of results with a suitable example.

Answer

1. Definition

A/B testing is an experiment with **two groups** used to determine which of two **treatments, products, procedures, or alternatives** is superior.

Usually, one option is the existing standard and is called the **control**, while the other is the new option being tested.

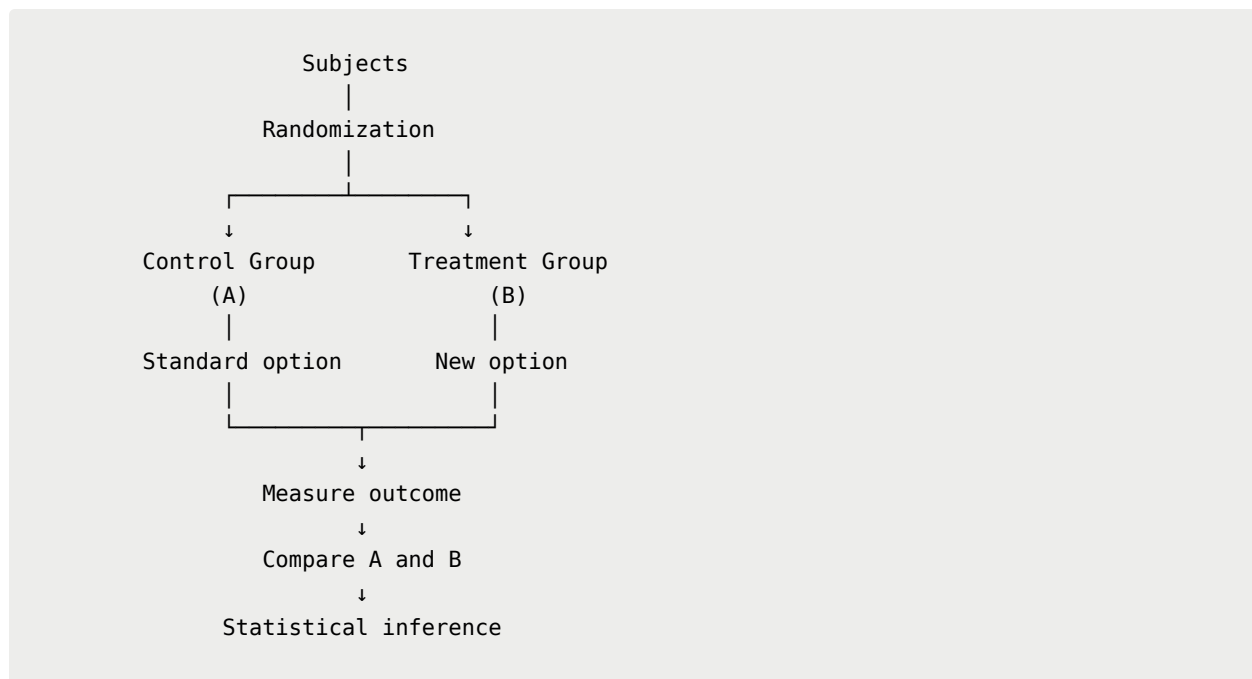
A/B testing is widely used in **web design and marketing**, where outcomes such as clicks, purchases, conversions and visit duration can be measured.

2. Key Components of A/B Testing

Term	Meaning
Treatment	Something to which a subject is exposed, such as a price or web headline
Treatment group	Group exposed to a particular treatment
Control group	Group exposed to no treatment or the standard treatment
Randomization	Randomly assigning subjects to treatments
Subject	Item exposed to the treatment, such as a web visitor
Test statistic	Metric used to measure the treatment effect

3. Design of an A/B Test

The basic design is:



Step 1: Define the objective

Clearly specify what decision the experiment is intended to support.

For example:

| Does a new webpage design increase the number of clicks?

Step 2: Select treatments

- **A** = existing/default treatment → **control**
- **B** = new treatment → **treatment group**

The treatment could be a **webpage design, product price, headline, advertisement**, etc.

Step 3: Randomize subjects

Subjects are **randomly assigned** to A or B.

Randomization is important because it helps ensure that factors other than the treatment are distributed between the groups rather than systematically favoring one group.

Thus, differences in outcomes can be attributed more reliably to the treatment rather than to differences between the subjects.

Step 4: Measure the outcome

A suitable **test statistic or metric** is selected to measure the effect.

Possible metrics include:

- **Clicks**
- **Purchases**
- **Visit duration**
- **Number of pages visited**
- **Conversions.**

Importantly, the primary metric should be **decided before the experiment**. Selecting the metric after seeing the results can introduce **researcher bias**.

4. Control and Treatment Groups

Control Group — A

The control group receives the **existing standard treatment** or no treatment.
It provides the **baseline for comparison**.

Treatment Group — B

The treatment group receives the **new treatment** being evaluated.

The outcome of B is compared with A to determine whether the new treatment produces an improvement.

5. Statistical Interpretation

5. Statistical Interpretation

Suppose the measured outcome has means:

$$\bar{x}_A = \text{mean outcome for A}$$

$$\bar{x}_B = \text{mean outcome for B}$$

The observed difference is:

$$D = \bar{x}_B - \bar{x}_A$$

A hypothesis test can then determine whether this observed difference could reasonably have occurred through **random chance**.

Hypotheses

If the objective is to determine whether B is better than A:

$$H_0 : \mu_B \leq \mu_A$$

$$H_1 : \mu_B > \mu_A$$

This is a **one-way (one-tail) test**, because only improvement in the direction favoring B is relevant.

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Alternatively, if we want to determine whether the treatments are different in either direction:

$$H_0 : \mu_A = \mu_B$$

$$H_1 : \mu_A \neq \mu_B$$

This is a **two-way (two-tail) test**.

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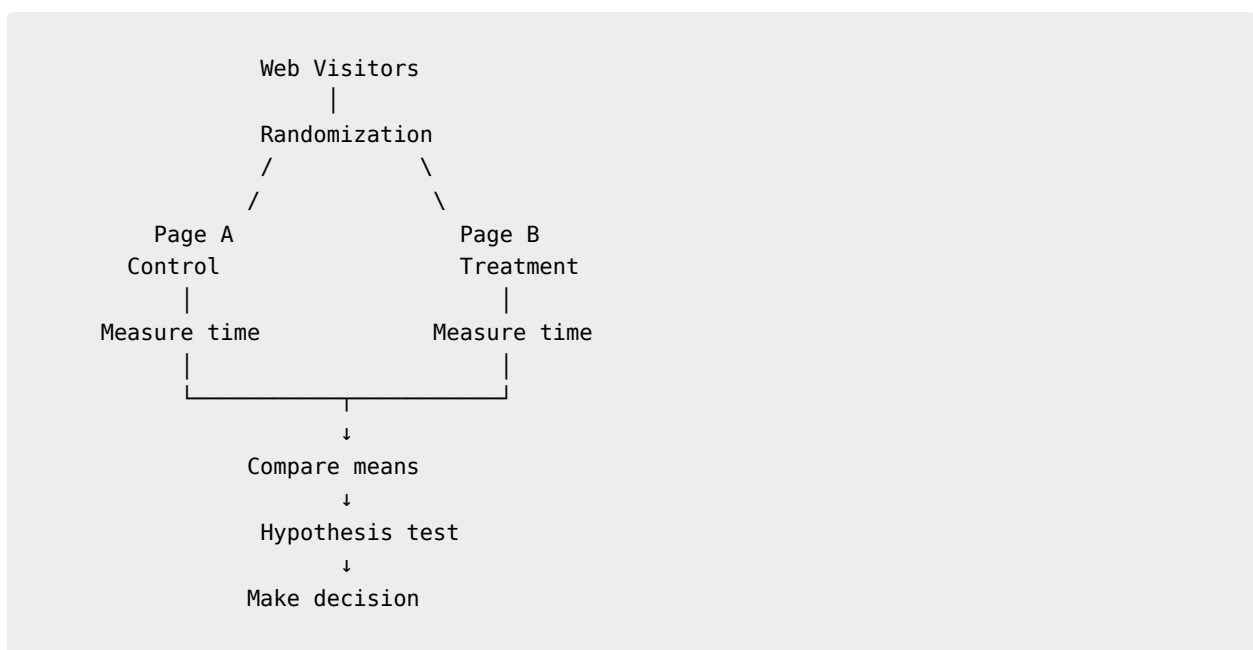
6. Example — Webpage A/B Test

Suppose an e-commerce company wants to test two webpage designs:

- **Page A:** Existing webpage → **Control**
- **Page B:** New webpage → **Treatment**

Visitors are randomly divided into the two groups.

Suppose the experiment measures **session duration**.



Suppose:

Suppose:

$$\bar{x}_A = 126.33 \text{ seconds}$$

$$\bar{x}_B = 162.00 \text{ seconds}$$

Therefore:

$$D = 162.00 - 126.33$$

$$D = 35.67 \text{ seconds}$$

Thus, Page B has an average session time **35.67 seconds greater** than Page A.

However, **a difference alone is not sufficient**. We must determine whether the difference is statistically significant.

A hypothesis test or permutation test can be used to assess whether such a difference could reasonably arise from **random chance**.

If the resulting **P-value is sufficiently small**, there is evidence against H_0 , supporting the conclusion that the treatment has an effect.

If the P-value is large, there is **insufficient evidence** to conclude that the observed difference is due to the treatment rather than random variation.

7. Importance of Randomization

Randomization is one of the most important principles of experimental design.

Without randomization, differences between the groups may be caused by **confounding factors** rather than the treatment.

For example, if Page B is shown mainly to experienced users while Page A is shown mainly to new users, a difference in session duration could be caused by the type of users rather than the webpage.

Therefore:

Randomization → Reduced bias → Fair comparison → More reliable inference

The notes emphasize that specifying a hypothesis and collecting data according to **randomization and random-sampling principles** helps protect against bias.

8. Advantages / Significance

1. Provides a **controlled comparison** between two alternatives.
2. **Randomization** reduces systematic differences between groups.
3. Allows the effect of a treatment to be measured using a predefined **test statistic**.

4. Statistical hypothesis testing helps distinguish genuine effects from **random chance**.
 5. Particularly useful in **web design, marketing and e-commerce**, where outcomes can be measured readily.
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Question 7:

Explain permutation tests in detail. Describe the procedure for conducting a permutation test and explain how it can be used to determine statistical significance.

Answer

1. Definition

A **permutation test** is a **resampling-based hypothesis test** used to determine whether an observed difference between two or more groups could reasonably be explained by **random chance**.

In a permutation test, the observations from the groups are **combined and randomly reallocated** to new groups. The statistic of interest is calculated repeatedly to create a **permutation distribution**.

It is also called a **randomization test**, **random permutation test**, or **exact test** in the exhaustive case.

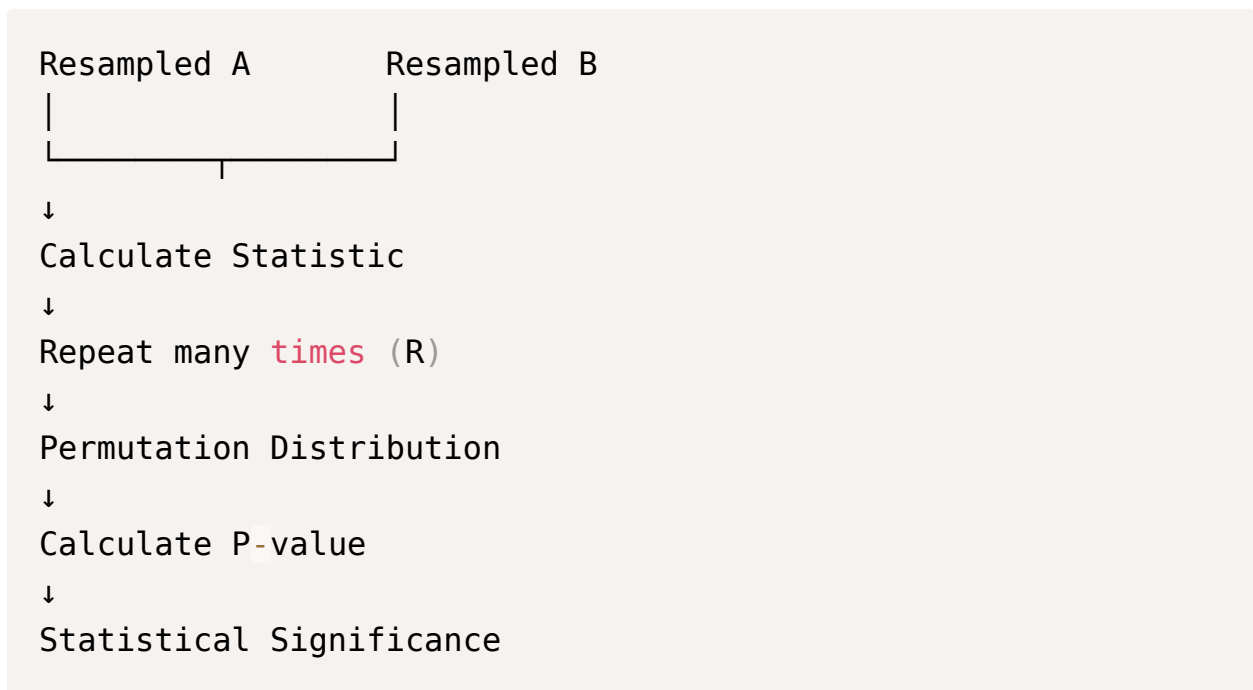
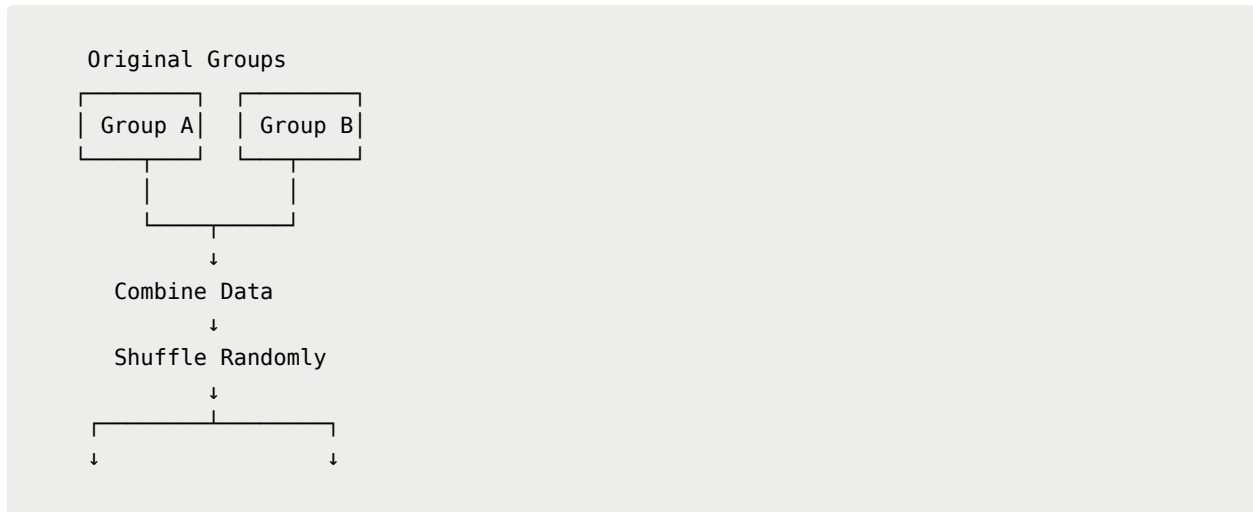
2. Basic Idea

Consider two groups, **A and B**.

The **null hypothesis** assumes that the treatments are equivalent, so the observations are **interchangeable** between the groups.

Therefore, if we combine the observations from A and B and randomly redistribute them into groups of the original sizes, the resulting differences represent the differences that could occur **by chance alone**.

Diagram



3. Step-by-Step Procedure

Step 1: State the hypotheses

Define the null and alternative hypotheses.

For example:

$$H_0 : \mu_A = \mu_B$$

$$H_1 : \mu_A \neq \mu_B$$

The null hypothesis represents the assumption that there is **no real difference between the treatments**.

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Step 2: Calculate the observed statistic

Choose a suitable **test statistic** and calculate it from the original groups.

For example, for two numerical groups:

$$D_{\text{obs}} = \bar{x}_A - \bar{x}_B$$

For binary data, the statistic could be the **difference in proportions**.

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Step 3: Combine the observations

Combine all observations from groups A and B into a **single data set**.

This combined data represents the **null model**, in which the groups are assumed to be equivalent and interchangeable.

Step 4: Shuffle and divide the data

Randomly shuffle the combined observations.

Then:

- Randomly select a sample having the **same size as Group A**.
- From the remaining observations, select a sample having the **same size as Group B**.
- The samples are drawn **without replacement**.

Step 5: Calculate the statistic

Calculate the same statistic for the newly created groups.

For example:

$$D_{\text{perm}} = \bar{x}_{A,\text{perm}} - \bar{x}_{B,\text{perm}}$$

Record this value.

This represents **one permutation iteration**.

Step 6: Repeat the procedure

Repeat the shuffling, division and calculation R times.

This produces:

$$D_1, D_2, D_3, \dots, D_R$$

These values form the **permutation distribution of the test statistic**.

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4. Determining Statistical Significance

The permutation distribution represents the amount of difference that can occur **under the null hypothesis** due to random chance.

We compare the observed statistic with this distribution.

P-value

The **P-value** is the proportion of permutation results that are **as extreme as or more extreme than the observed result**.  BAD702-module-3-textbook(1)

For a one-sided test:

$$p = \frac{\text{Number of permutation statistics as or more extreme than } D_{\text{obs}}}{R}$$

For example, suppose 10,000 permutations are performed and 300 produce a difference at least as large as the observed difference:

$$p = \frac{300}{10000} = 0.03$$

If:

$$\alpha = 0.05$$

then:

$$0.03 < 0.05$$

Therefore, **reject** H_0 and conclude that the observed difference is statistically significant.

5. Interpretation

Small P-value

$$p \leq \alpha$$

A small P-value means that the observed result would be **unusual under the null hypothesis**.

Hence, there is evidence against H_0 .

Large P-value

$$p > \alpha$$

A large P-value means that a result as extreme as the observed result can occur **reasonably often through random chance**.

Therefore, there is **insufficient evidence to reject H_0** .

The P-value does not prove that the null hypothesis is true.

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6. Example — A/B Test

Suppose an e-commerce company tests two prices, **A and B**, and compares their conversion rates.

The observed difference in conversion rates is:

$$D_{\text{obs}} = 0.0368$$

Under the null hypothesis, assume both prices have the **same conversion rate**.

The conversion results from both groups are combined and repeatedly shuffled. For each permutation, the difference in conversion rates is calculated.

Suppose the permutation test gives:

$$p = 0.308$$

This means that a difference as large as the observed difference, or larger, could occur by random chance about **30.8% of the time** under the null model.

Since:

$$0.308 > 0.05$$

we **fail to reject H_0** .

Therefore, the observed difference is **not statistically significant at the 5% level**.

7. Types of Permutation Tests

A. Random Permutation Test

The data is randomly shuffled and divided repeatedly.

With a sufficiently large number of repetitions, its results approach those of the exhaustive permutation test.

B. Exhaustive Permutation Test

All possible ways of dividing the combined data into groups are considered.

- Provides an **exact test**.
- Practical mainly for **small sample sizes** because the number of possible permutations becomes very large.

C. Bootstrap Permutation Test

The resampling is performed **with replacement**, unlike the ordinary permutation procedure.

It models both:

- Random assignment of treatments.
 - Random selection of subjects from a population.
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8. Advantages

1. **Simple to understand and implement.**
 2. Does not require assumptions about **normally distributed data**.
 3. Can handle **numeric or binary data**.
 4. Can be used when sample sizes are **equal or different**.
 5. Provides a direct way to investigate the role of **random variation**.
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Question 8:

Explain statistical significance and practical significance. Discuss why statistical significance alone may not be sufficient for making decisions.

Answer

1. Statistical Significance

Statistical significance is a measure used to determine whether an observed result is **more extreme than what chance variation would reasonably produce** under a specified chance model or null hypothesis.

It is commonly assessed using a **P-value** and a predetermined significance level, called **alpha (α)**.

P-value

The P-value is the probability, under the null hypothesis, of obtaining a result **as unusual or more extreme than the observed result**.

A commonly used significance level is:

$$\alpha = 0.05$$

Decision rule:

$$p \leq \alpha \Rightarrow \text{Statistically significant}$$

$$p > \alpha \Rightarrow \text{Not statistically significant}$$

A statistically significant result indicates that the observed effect is unlikely to be explained by ordinary chance variation under the assumed model.

2. Practical Significance

Practical significance refers to whether the observed effect is **large enough to have meaningful real-world consequences**.

A result can be statistically significant but have **little or no practical importance**. Large samples can make very small differences statistically significant because they provide enough evidence to rule out chance as an explanation.

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Therefore:

$$\text{Statistical significance} \neq \text{Practical importance}$$

3. Example

Suppose a company introduces a new webpage.

- Old webpage average response time = **10.00 seconds**
- New webpage average response time = **9.99 seconds**

The difference is:

$$D = 9.99 - 10.00 = -0.01 \text{ seconds}$$

Assume a very large sample produces:

$$p < 0.05$$

Therefore, the difference is **statistically significant**.

However, an improvement of only **0.01 seconds** may have essentially no meaningful effect on users or business performance.

Hence:

- **Statistically significant:** Yes
- **Practically significant:** Possibly no

This illustrates why ruling out chance does not automatically make an effect important.

4. Why Statistical Significance Alone Is Not Sufficient

1. It does not measure effect size

A P-value or statistical significance tells us whether the result is unusual under the null model, but it **does not measure the size of the effect or its importance**.

For example, a difference of:

0.001

may be statistically significant with a huge sample but still be irrelevant in practice.

2. Large samples can make tiny effects significant

As sample size increases, estimates become more precise. Consequently, even a **small, non-meaningful difference** may produce a very small P-value.

Thus:

Large sample → small detectable difference → statistical significance

But this does not imply that the difference is useful in practice.

3. A non-significant result does not prove no effect

If:

$p > 0.05$

we generally **fail to reject the null hypothesis**.

This does not prove that the effect is zero. The sample may simply be too small to detect the effect.

Thus, a non-significant result should be interpreted as **"effect not proven"**, rather than **"no effect exists."**

4. P-value is not the probability that the hypothesis is true

A common misconception is:

$$p = P(H_0 \text{ is true})$$

This is incorrect.

The P-value represents the probability of obtaining results as extreme as the observed results **given the chance model/null hypothesis**, not the probability that the hypothesis itself is true.

5. Business and scientific decisions require more information

The source emphasizes that **scientific conclusions and business or policy decisions should not be based only on whether a P-value passes a particular threshold.**

A decision should also consider:

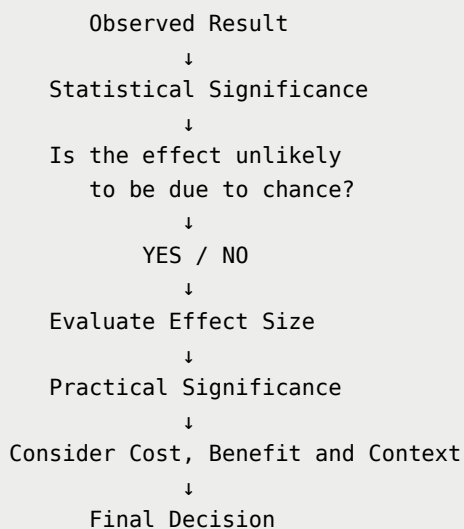
- **Magnitude of the effect**
 - **Practical consequences**
 - **Cost and benefit**
 - **Uncertainty and variability**
 - **Sample size**
 - **Confidence intervals**
 - The context of the problem.
-

5. Statistical vs Practical Significance

Aspect	Statistical Significance	Practical Significance
Meaning	Result is unlikely under the chance/null model	Effect is meaningful or useful in the real world
Main measure	P-value and α	Effect size and real-world impact
Depends strongly on	Sample size and variability	Magnitude and consequences of effect
Main question	"Could this result reasonably occur by chance?"	"Is this difference important enough to matter?"
Large sample effect	Tiny effects may become significant	Tiny effects may still be unimportant

6. Relationship

The correct decision-making framework is:



Therefore, **statistical significance is only one piece of information** in a decision.

7. Important Point from Data Science

For a data scientist, a P-value can be a **useful metric** for determining whether an apparently interesting result lies within the range of normal chance variability.

However, it should be treated as **another point of information rather than the controlling decision criterion**.

Question 9:

Explain Type I and Type II errors in hypothesis testing. Discuss their relationship with statistical significance and decision-making.

Answer

1. Introduction

In **hypothesis testing**, a decision is made about a **null hypothesis H_0** based on sample data. Since the decision is based on a sample rather than the entire population, errors are possible.

There are two important types of errors:

- **Type I error** — concluding that an effect is real when it is actually due to chance.
- **Type II error** — concluding that an effect is due to chance when it is actually real.

2. Type I Error

A **Type I error** occurs when we **reject the null hypothesis H_0** even though it is actually true.

In simple terms:

Type I error = False positive

That is, we mistakenly conclude that a statistically significant effect exists when the observed result is actually due to **random chance**.

Example

Suppose an A/B test compares two webpages.

H_0 : There is no difference between A and B

If we reject H_0 and conclude that Page B is better, when in reality there is **no real difference**, a Type I error has occurred.

3. Type II Error

A **Type II error** occurs when we **fail to reject H_0** even though a real effect actually exists.

In simple terms:

Type II error = False negative

We incorrectly conclude that the observed effect is due to chance when it is actually a **real effect**.

Example

Suppose a new drug genuinely improves patient recovery.

If a statistical test fails to detect this improvement and concludes that there is insufficient evidence of an effect, a **Type II error** has occurred.

The notes emphasize that in practice, a non-significant result is often better interpreted as **"effect not proven"**, because a larger sample might produce sufficient evidence to detect the effect.

4. Relationship with Statistical Significance

Statistical significance is used to determine whether an observed result is more extreme than what could reasonably be produced by chance under the null hypothesis.

The **P-value** is:

$P(\text{result as extreme as observed} \mid H_0)$

A significance level α , commonly 0.05, is selected as the threshold.

Decision rule

$$p \leq \alpha \Rightarrow \text{Reject } H_0$$

$p > \alpha \Rightarrow$ Fail to reject H_0

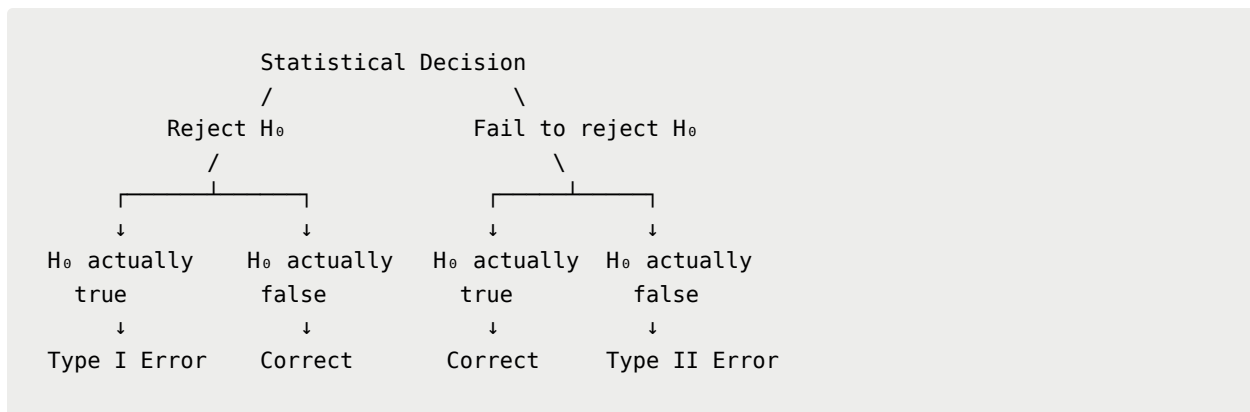
If we reject H_0 when it is actually true, we make a **Type I error**.

If we fail to reject H_0 when a real effect exists, we make a **Type II error**.

5. Decision Table

Reality	Decision: Reject H_0	Decision: Fail to reject H_0
H_0 is true	Type I error	Correct decision
Real effect exists	Correct decision	Type II error

This can be represented as:



6. Type I Error and Alpha

The significance level α controls the probability threshold for a Type I error.

For example:

$$\alpha = 0.05$$

means that the testing procedure uses a **5% threshold of unusualness** under the null hypothesis for declaring a result statistically significant.

Therefore:

$\alpha \downarrow \Rightarrow$ more stringent evidence required

A smaller significance level reduces the tendency to make Type I errors, but can make it harder to detect real effects.

7. Type II Error and Sample Size

Type II errors are strongly related to the **ability of a study to detect a real effect**.

If the sample size is too small, a genuine effect may not produce a sufficiently small P-value.

Thus:

Small sample → low ability to detect effect → greater risk of Type II error

Increasing sample size generally provides more information and makes it easier to detect meaningful effects.

Therefore, a non-significant result does **not** prove that no effect exists. It may simply mean that the available sample was insufficient to establish the effect.

8. Effect on Decision-Making

Both errors can have important consequences.

Type I error — False positive

```
No real effect
  ↓
Reject H0
  ↓
Claim effect exists
  ↓
Wrong action
```

For example, a company may adopt a new webpage because an A/B test incorrectly indicates that it performs better.

Type II error — False negative

Real effect exists
↓
Fail to reject H_0
↓
Effect not detected
↓
Miss useful opportunity

For example, a company may reject an actually effective new webpage because the experiment did not have enough evidence to detect its improvement.

Therefore, decision-making must consider the **consequences of both types of errors**.

9. Important Limitation of Statistical Significance

Statistical significance alone should **not control the final decision**.

A P-value does not measure:

- The **size of the effect**
- The **practical importance** of the result
- The probability that the hypothesis is true.

For example, a very small effect can become statistically significant with a sufficiently large sample.

Therefore, decisions should also consider:

- **Effect size**
- **Sample size**
- **Practical significance**
- **Costs and benefits**
- **Consequences of Type I and Type II errors**
- **Confidence intervals and uncertainty**

The notes specifically emphasize that P-values should be treated as **one point of information**, rather than the sole controlling factor in a decision.

10. Summary

Error	Meaning	Common Name	Example
Type I	Reject true H0	False positive	Claiming a treatment works when it does not
Type II	Fail to reject false H0	False negative	Missing a treatment effect that actually exists

Question 10:

Explain resampling methods used in statistical inference and compare permutation tests with bootstrap methods.

Answer

1. Definition of Resampling

Resampling in statistics means repeatedly sampling values from **observed data**, with the general goal of assessing the **random variability of a statistic**. It can also be used to assess and improve the accuracy of some machine-learning models.

The two main resampling procedures are:

1. **Bootstrap**
2. **Permutation tests**

The **bootstrap** is mainly used to assess the **reliability and variability of an estimate**, whereas **permutation tests** are mainly used to **test hypotheses**, typically involving two or more groups.

2. Bootstrap Method

A **bootstrap sample** is obtained by sampling **with replacement** from the observed data.

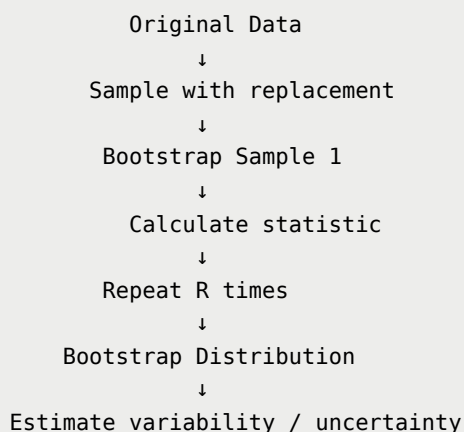
Conceptually, the original sample is treated as a hypothetical population containing the information available in the sample. Repeated samples are then drawn from it to estimate the sampling distribution of a statistic.

Procedure

For a sample of size n :

1. Draw one observation and **record it**.
2. Replace it before the next draw.
3. Repeat until n observations have been obtained.
4. Calculate the statistic, such as the **mean**.
5. Repeat the process R times.
6. Use the resulting R statistics to estimate the **sampling distribution and variability**.

Diagram



The bootstrap is useful because it can estimate sampling distributions even when a convenient mathematical approximation is unavailable.

3. Permutation Test

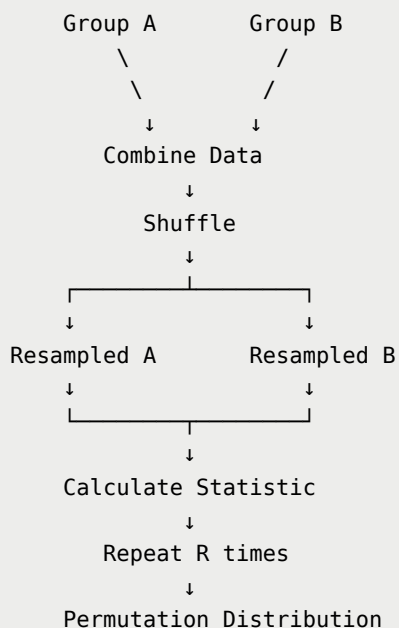
A **permutation test** is a resampling procedure primarily used for **hypothesis testing**.

For two groups, the observations are combined and then randomly reallocated to groups of the original sizes. This represents the **null hypothesis** that the treatments do not differ.

Procedure

1. **Combine** observations from groups A and B.
2. **Shuffle** the combined data.
3. Randomly draw, **without replacement**, a sample of the same size as group A.
4. Draw the remaining observations for group B.
5. Calculate the same statistic used for the original groups.
6. Repeat the procedure R times.
7. The resulting statistics form the **permutation distribution**.

Diagram



↓
Hypothesis Test

The P-value is obtained by determining how often the permutation statistic is **as extreme as or more extreme than the observed statistic**.

4. Comparison: Bootstrap vs Permutation Test

Feature	Bootstrap	Permutation Test
Main purpose	Estimate variability/reliability of a statistic	Test a hypothesis
Sampling	With replacement	Generally without replacement
Basic idea	Resample from the observed data	Shuffle and reallocate observations between groups
Null hypothesis	Not necessarily imposed	Explicitly represents the null hypothesis
Groups	Usually based on a sample or population	Typically involves two or more groups
Output	Bootstrap distribution	Permutation distribution
Main use	Standard error, confidence intervals, uncertainty	P-values and statistical significance
Example	Estimate variability of a sample mean	Test whether two group means differ

The distinction is important: **bootstrap always implies sampling with replacement**, while the broader term **resampling** can include both bootstrap and permutation procedures.

5. Important Difference in Sampling

Bootstrap

Sampling with replacement\boxed{\text{Sampling with replacement}}

An observation can therefore appear **multiple times** in the same bootstrap sample.

Example:

```
Original:      A  B  C  D  E
Bootstrap:     B  D  B  A  E
               ↑
             B selected twice
```

Permutation

Reallocation without replacement\boxed{\text{Reallocation without replacement}}

Each observation remains available only once in a particular permutation.

```
Original:      A  B  C  D  E  F
Permutation:   D  A  F | C  E  B
               Group A   Group B
```

6. Use in Statistical Inference

Resampling provides a way to construct a **reference or sampling distribution directly from the observed data**, rather than relying entirely on formula-based theoretical distributions.

Bootstrap

Used to answer:

┃ **How variable is my estimate?**

For example:

- Variability of a mean
- Reliability of an estimate
- Confidence intervals

Permutation test

Used to answer:

| **Could the observed difference have occurred through random chance?**

For example:

- Compare two group means
- A/B testing
- Test differences in proportions

Permutation tests are particularly flexible because the data can be **numeric or binary**, sample sizes can be the same or different, and assumptions of normally distributed data are not required.

7. Example

Suppose an A/B test compares session times for two webpages.

Bootstrap approach

We want to estimate the **variability of the average session time**.

Repeatedly sample session times **with replacement**, calculate the mean each time, and form a bootstrap distribution of means.

This distribution can be used to assess the **uncertainty of the estimated mean**.

Permutation approach

We want to determine whether Page A and Page B have genuinely different session times.

Combine all session times, repeatedly **shuffle and divide** them into groups having the original sizes, and calculate the difference in means each time.

If the observed difference is far into the tail of the permutation distribution, it provides evidence against the null hypothesis.
